

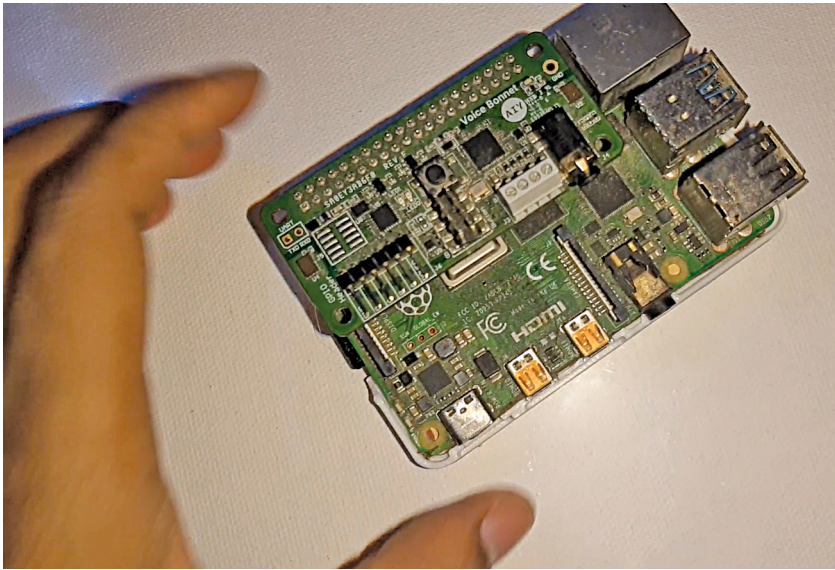


Figure 1: Setting the Voice bonnet

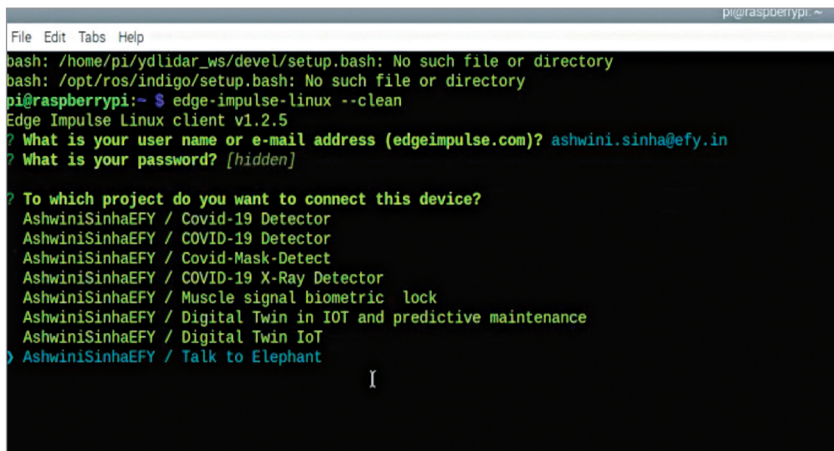


Figure 2: Connecting RPI with project

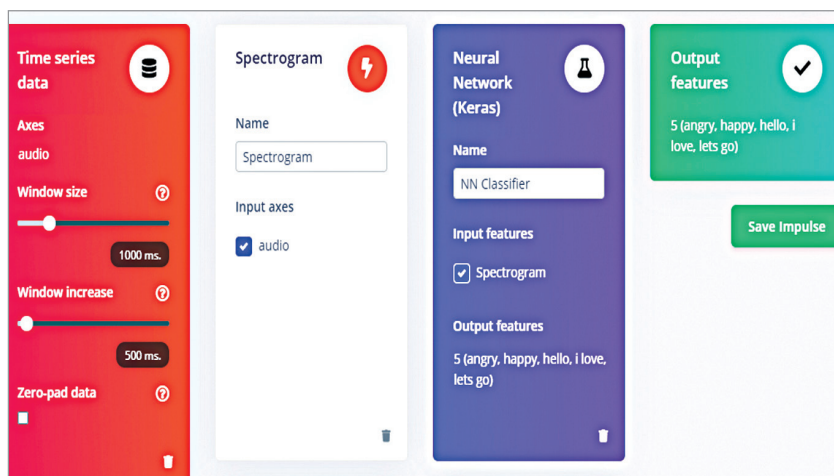


Figure 3: Setting processing blocks

following commands, create a new project named Fauna Translator:

```
curl -sL https://deb.nodesource.com/setup_12.x | sudo bash -
```

```
sudo apt install -y gcc g++ make build-essential nodejs sox gstreamer1.0-tools gstreamer1.0-plugins-good gstreamer1.0-plugins-base gstreamer1.0-plugins-apps
```

```
npm config set user root && sudo npm install edge-impulse-linux -g --unsafe-perm
```

After this, connect the Raspberry Pi project with Edge Impulse using Edge-impulse-linux.

Next, open the terminal and select the project name, after which you are given a URL for feeding the datasets. In that URL, enter the captured sounds of animals and appropriately label them based on different emotions (angry, glad, hungry) or expressions (“Let’s go”, “I love you”).

Note: Before proceeding with sound data capture, put the AIY Voice bonnet onto the Raspberry Pi as shown in Figure 1.

ML model

Select the learning and processing blocks for training the ML model. Here we use Spectrogram for the processing block and Keras for the learning block. Using these, extract different audio parameters for training the ML model to learn from datasets. Then test the model and keep refining it until you are satisfied with its accuracy.

To deploy the ML model, go to the deploy option and select *Linux board*. Install it and clone the SDK of edge impulse.